

abtestkit

```
library(abtestkit)
```

What this package does

LINK TO GITHUB: <https://github.com/mbsperry-school/abtestkit>

abtestkit is a toolkit for planning and analyzing binary-outcome A/B tests. It covers both frequentist and Bayesian approaches side by side, aimed at someone who understands what an A/B test is but has never worked with Bayesian methods.

There are two phases: planning (before the experiment) and analysis (after the experiment). This vignette walks through both using a real marketing dataset.

The dataset

The data comes from a Kaggle marketing A/B test where 588,101 users were exposed to either an advertisement or a public service announcement (PSA) placeholder. The outcome is whether the user converted (made a purchase).

We split this into two pieces:

- **historical_data**: 1,000 PSA users treated as pre-experiment baseline data. This is what we “know” before the experiment starts.
- **experiment_data**: 10,000 users per arm sampled from the remaining data. This is the actual experiment.

```
data(historical_data)
data(experiment_data)
```

Phase 1: Pre-experiment planning

Before running an experiment, you need to answer two questions: how many users do I need, and what do I believe going in?

The frequentist side answers the first question with a sample size calculation. The Bayesian side answers the second by turning your historical data into a prior distribution.

`pre_experiment_comparison` handles both at once. You pass in your historical data, the column with the binary outcome, and the minimum detectable effect (the smallest change worth detecting). Here we set `mde = 0.01`, meaning we want to detect a 1 percentage point lift.

```
mde <- 0.01
outcome_column <- "converted"
group_column <- "test.group"
```

```

control_value <- "psa"
treatment_value <- "ad"

pre_experiment_data <- pre_experiment_comparison(
  mde = mde,
  data = historical_data,
  outcome_column_name = outcome_column
)

```

The function returns two things: the sample size needed per group (frequentist) and a Beta prior (Bayesian).

```

n <- pre_experiment_data$n
experiment_prior <- pre_experiment_data$experiment_prior

cat("Required sample size per group:", n, "\n")
#> Required sample size per group: 4569
cat("Prior alpha:", round(experiment_prior$alpha, 3), "\n")
#> Prior alpha: 0.354
cat("Prior beta:", round(experiment_prior$beta, 3), "\n")
#> Prior beta: 10.73
cat("Prior method:", experiment_prior$method, "\n")
#> Prior method: quantile

```

The sample size tells you how long to run the test. The prior encodes what you knew about conversion before seeing any experiment data. Because our historical conversion rate is low (well below 50%), the prior was fit using quantile matching, which handles skewed distributions better than moment matching.

Phase 2: Post-experiment analysis

Now we have experiment results. `post_experiment_comparison` runs both analyses on the same data. The frequentist side runs a two-proportion z-test. The Bayesian side updates the prior with each group's observed conversions and compares the resulting posterior distributions.

```

post_experiment_data <- post_experiment_comparison(
  data = experiment_data,
  outcome_column_name = outcome_column,
  group_column_name = group_column,
  control_value = control_value,
  treatment_value = treatment_value,
  n = n,
  experiment_prior = experiment_prior
)

```

Results

`full_comparison` puts everything together: side-by-side visuals and a plain-language interpretation of what both methods found.

```

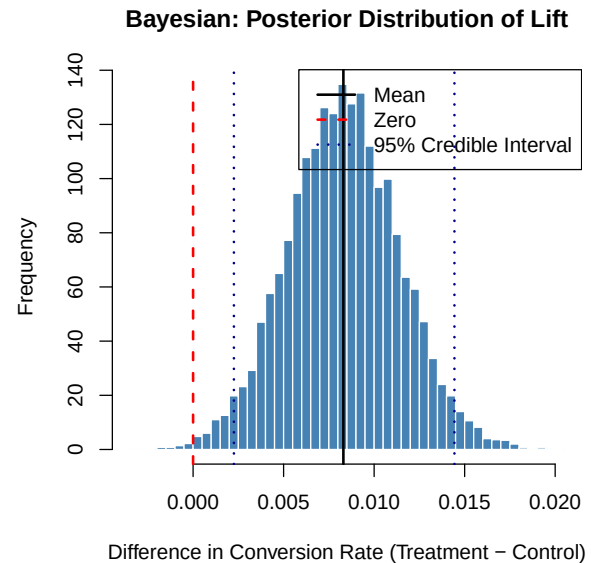
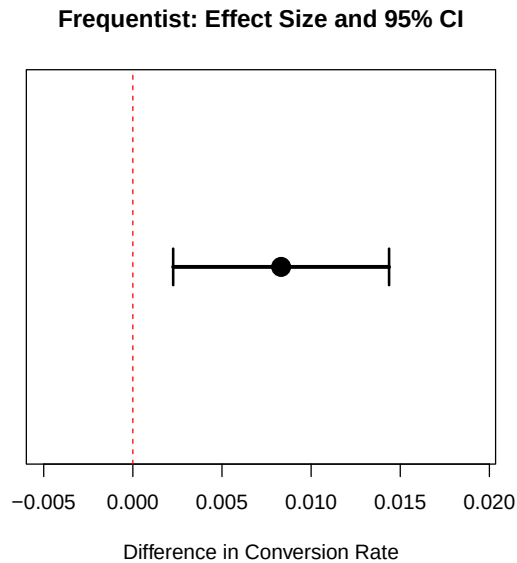
full_comparison(
  freq_results = post_experiment_data$freq_results,
  bayes_results = post_experiment_data$bayes_results,

```

```

posterior_control = post_experiment_data$posterior_control,
posterior_treatment = post_experiment_data$posterior_treatment
)

```



```

#> === A/B Test Results Interpretation ===
#>
#> -- Frequentist Results --
#> Observed effect: 0.0083
#> p-value: 0.0071
#> 95% Confidence Interval: [ 0.0023 , 0.0144 ]
#> The frequentist confidence interval says: if we repeated this
#> experiment many times, 95% of the intervals we compute would
#> contain the true effect.
#>
#> The result IS statistically significant at alpha = 0.05
#> This means: if there were truly no difference between groups,
#> we would see data this extreme less than 5 % of the time.
#> We reject the null hypothesis that the groups are the same.
#>
#> -- Bayesian Results --
#> P(treatment > control): 0.996
#> Expected lift: 0.0083
#> 95% Credible Interval for lift: [ 0.0023 , 0.0144 ]
#> The Bayesian credible interval says: given our prior beliefs and
#> the observed data, there is a 95% probability that the true
#> effect falls inside this interval.
#>
#> The Bayesian analysis gives strong evidence that treatment is
#> better.
#> There is a 99.6 % probability that the treatment group has a higher conversion rate.
#>
#> -- Do the two methods agree? --
#>

```

```
#> Yes. Both methods agree that the treatment group performed better.  
#> The frequentist test found a significant positive effect, and the  
#> Bayesian analysis assigns a high probability to treatment being  
#> better.
```

The left plot shows the frequentist confidence interval for the difference in conversion rates. If the interval excludes zero (the red dashed line), the result is statistically significant.

The right plot shows the Bayesian posterior distribution of the lift. The black line is the expected lift, the dashed blue lines mark the 95% credible interval, and the red dashed line is zero. If most of the distribution sits to the right of zero, the Bayesian analysis favors the treatment.

The printed interpretation below the plots explains what each method concluded, whether they agree, and the difference between a confidence interval and a credible interval.